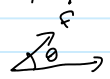
In Class notes

Thursday, October 16, 2025 9:07 AM

- Work and energy are inter connected
- work - energy theorem
  - ↳ Lets us make formulas for energy
- Mathematically, were learning how to multiply vectors
- work is a scalar quantity but it can have positive and negative signs,
- A force causes displacement and when this happens, work has been done
- The formula for this is $W = \vec{F} \cdot \Delta d$
- if the force is constant, $W = \vec{F} \cdot \Delta d \cos\theta$
- If not constant, use calculus, graph
- This formula comes from definition

[Textbook definition: The product of the magnitude of an object's displacement and the component of the applied force in the direction of the displacement]

- Dot product [one of many ways to multiply without scalars]
- $W_{[\text{done by any force}]} = |\vec{F}_{any}||\vec{\Delta d}| \cos\theta$



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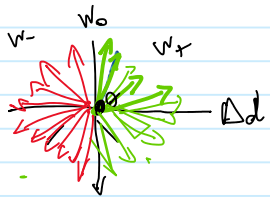
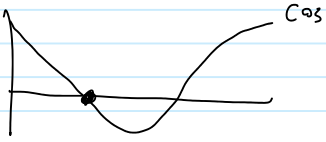


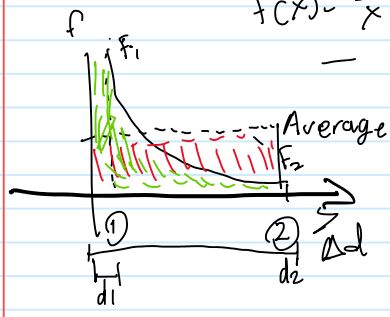
- If force angle is pointing in the green, you get positive work
- If force angle is pointing in the red, you get negative force
- If the force is negative or positive is dependent on the angle, not E, N, S, W.



- For uniform circular motion, no work is being done
- (-) work represents the loss of energy
- for a cos of 90°, you get no work being done.

- for a f - d graph regarding non-constant force:
  - ↳ If sum of the area is positive, the work is positive
    - ↳ vice versa
  - ↳ for graphs, where we can't calculate area, we can

$f(x) = \frac{1}{x}$

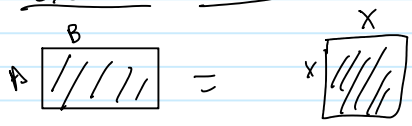


$= \sqrt{f_1 f_2}$

Average force $= \sqrt{\frac{1}{d_1}\left(\frac{1}{d_2}\right)}$

Explaination

Geometric mean = mean of f



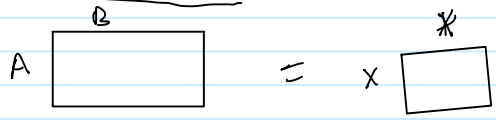
$AB = x^2$

$x = \sqrt{AB}$

$W = f_{avg} \Delta d$

$= \sqrt{f_1 f_2} \Delta d$

Arithmic mean : Add



$P = 2(A+B)$

$P = 4x$

$x = \frac{a+b}{2}$

↳ This what we use for averaging marks

Personal Textbook addons:

- Joules = Nom or $kg\,m^2/s^2$
- Negative work happens when forces and displacement are in opposite directions.
- No displacement = no work
- for problems with uniform motion, work will only be done when there is a force and acceleration.
  - ↳ Ex: Example 3 in 4.1

- No work is done when
  - ↳ No force
  - ↳ $\theta = 90$
  - ↳ No displacement.

- Review Practice Tutorial 3 4.1 for uniform motion and work.

- A frictional force acts and does work on any surfaces that slide past each other